

An Improved Exact Computer-Assisted Lower Bound for Packing 17 Unit Squares in a Square: $s(17) > 4.6140071$

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Abstract

Let $s(17)$ be the least side length of a square containing seventeen pairwise interior-disjoint unit squares with arbitrary orientations. We give an exact computer-assisted certificate proving $s(17) \geq 4.614007067908891248924345449337\dots$, and the strict inequality $s(17) > 4.6140071$. The certificate is a nonnegative rational measure on 387 points of total mass $16.99990022 < 17$ that assigns mass at least 1.000222750 to every contained unit square, verified exactly at 92,161 rational directions. The previous best certified bound was $s(17) > 4.613028635886$ (Mira-acc, September 8, 2026); the improvement is $+9.8 \times 10^{-4}$ and is obtained by re-optimizing Mira-acc's own measure along two axes they did not explore: the atom-core side B (smaller cores trade coverage for dilation gain) and the direction-net density m (denser nets halve the dilation defect). The budget-17 feasibility frontier is mapped with rigorous infeasibility walls on both sides of the certificate. All computations use exact rational arithmetic; the certificate is replayed and validated by three independent engines, including Mira-acc's own exact kernel in both of its accumulation variants.

1 Result and provenance

Write $Q_L = [0, L]^2$. Squares in a packing may touch but their interiors must be disjoint.

Theorem 1. *Let*

$$L = \frac{4613105}{1000000}, \quad B = \frac{9998}{10000}, \quad h = \frac{207107}{46080000000}.$$

Then

$$s(17) \geq S_* := \frac{L\sqrt{1+h^2}}{B(1+h)} = 4.614007067908891248924345449337\dots,$$

and $s(17) > a$ for every rational $a < S_$; in particular*

$$s(17) > 4.6140071.$$

Remark 2 (The open bracket). The best known packing remains Bidwell's 1998 construction [5, 6] of side $4.67553009360455\dots$. Thus

$$4.6140071 < s(17) \leq 4.67553009360455\dots$$

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Remark 3 (Provenance and honest lineage). This work continues the public 2026 line of weighted-atom certificates: Burns [1] proposed the scheme (4.4811); Massaccesi [2] introduced the exact rational verifier architecture (4.5058); Levy [3] added the dilation corollary used here (4.59); Mira-acc [4] converged a 1,620-atom measure and certified 4.613028635886 with it (September 8, 2026). **Our measure was built starting from Mira-acc’s 1,620-atom measure; without their work this result would not exist.** Our contribution is the re-optimization of the measure and the full extraction of its value along the (L, B, m) frontier described below, together with the verification machinery and the frontier’s rigorous infeasibility walls.

2 The framework we use

We recall the standard weighted-atom argument [1, 2, 3, 4], stated in the form our certificate uses.

Definition 4 (cores and the direction net). A B -core of a unit square Q at orientation θ is the closed square concentric with Q of side B , at the same orientation. The *direction net* at resolution m is the set of rational directions

$$(c, s) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right), \quad t = \frac{k}{m} T, \quad k = 0, \dots, m, \quad T = \frac{207107}{500000},$$

covering $[0, \pi/4]$ with tangent half-gap at most $h = T/m$.

Lemma 5 (net reduction). *Every unit square $Q \subset Q_L$ contains, concentrically, a closed B -core at one of the net directions, and that core lies strictly inside the open unit square whenever $B^2(1+h)^2 < 1 + h^2$.*

Proof sketch. The orientation θ of Q differs from some net direction by at most h in tangent, hence by $\varepsilon < h$ in angle; rotating the B -core about the common center to the net direction scales its bounding box by at most $\cos \varepsilon + \sin \varepsilon \leq 1 + h$, and $B(1+h)$ -cores fit inside Q strictly by the stated inequality. $\square$

Definition 6 (atoms and capture). An *atom* is a rational point $a_i \in Q_L$ with rational weight $w_i > 0$; atom a_i *covers* the placements p (core centers, in the core’s rotated coordinate frame) with $\|p - a_i\|_\infty \leq B/2$. A placement p is *captured* if the total weight of atoms covering it is at least 1.

Lemma 7 (capture lemma). *Suppose every feasible placement at every net direction is captured, and the total atom weight is $W < 17$. Then no packing of 17 unit squares exists in Q_L ; hence $s(17) \geq L$.*

Proof sketch. Seventeen interior-disjoint unit squares have interior-disjoint B -cores. Each atom’s B -square can fully contain at most one such core: two contained core centers would lie within B of each other in L^∞ , forcing the two cores to overlap. So the atoms’ total weight is at least the number of cores, $17 > W$ — contradiction. The minimum captured weight in our certificate is 1.000222750, bounded away from 1, which absorbs the standard boundary-shrink argument [5]. $\square$

Corollary 8 (dilation, Levy/Mira-acc form). *A certificate at (L, B, m) with strict containment $B^2(1+h)^2 < 1 + h^2$ implies $s(17) \geq S_* = L\sqrt{1+h^2}/(B(1+h))$, and $s(17) > a$ for every rational $a < S_*$.*

3 The certificate

Table 1 summarizes the certificate. Section 6 describes its three-engine verification.

field	value
container side L	$4613105/1000000 = 4.613105$
core side B	$9998/10000 = 0.9998$
direction net	$m = 92160$ (92,161 directions; $h = 207107/4608000000$)
atoms	387, exact rational coordinates and weights
total weight	$16999900220/1000000000 = 16.99990022 < 17$
minimum capture	$1000222750/1000000000 (= 1.000222750)$
strict containment	$B^2(1+h)^2 = 0.9996076 \dots < 1 \checkmark$
S_* (dilation)	$4.614007067908891248924345449337 \dots$

Table 1: The certificate (file `certificate/cert_4613105_B9998_k92160.json`).

4 Building the measure

4.1 Starting point: re-extracting the incumbent’s claim

The seed is Mira-acc’s 1,620-atom measure [4]. Two observations launched the campaign:

- (i) *Their published claim is conservative.* Re-verified by our independent integer engine, the same measure passes at $L = 4.613035$ and $L = 4.613045$: the binding direction’s capture (1.0000021) is insensitive to L at that scale.
- (ii) *The bound is dominated by the dilation defect h , not by L .* Since $S_* \approx (L/B)(1-h)$, halving h (denser net) is worth more than squeezing L — and shrinking B gains a full factor of B^{-1} if the measure can be repaired to cover at the smaller core side.

4.2 Scan-driven row generation

Feasibility at each candidate (L, B) is explored with a purpose-built row-generation loop:

- *Separation oracle.* For each net direction the feasible u -range is split into 32 bands and the lowest-charge cells per band (up to 4, from distinct slabs) are harvested — spatially distributed rows, so one bad cell cannot shield its neighbours.
- *Master LP.* Maximize the minimum charge λ at fixed total budget $16.9999 < 17$ (HiGHS interior point; dual simplex was $7\times$ slower and degenerate). A min-total variant over row subsets gives a rigorous lower bound on the budget any full-coverage measure needs — the feasibility oracle used for the walls below.
- *Support growth, then freeze.* Atoms are added at worst-violated placements (growth phase) and then frozen (settle phase): growth-induced oscillations were breaking settled regions.
- *Dense-net endgame.* Harvest on the same $m = 5760$ net used for certification: a measure clean on a $4\times$ coarser net failed exact verification at 0.979 — the inter-net gap is about 2%.
- *Exact positions.* Atom coordinates are emitted as exact float64 rationals (denominator snapping once cost 0.012 of charge); weights are ceiling-rationalized to 10^9 , which only increases charges.

4.3 The two axes

The B -axis. At $L = 4.61306$, certified measures were built at $B = 0.9999$ ($S_* = 4.6131896$), $B = 0.99985$ (4.6134203), and $B = 0.9998$ (4.6136510). The $B = 0.9998$ measure turned out to oversatisfy its target: the same 387-atom measure also certifies $L = 4.61308$, 4.61309, 4.6131 and 4.613105.

The m -axis. The certified measure at $(L, B) = (4.613105, 0.9998)$ was re-verified on successively denser nets. Its minimum capture is *identical* on every net (1.000222750 — the binding direction has a genuine continuous margin), so denser nets pass for free:

m	directions	S_*
5,760	5,761	4.6136960
11,520	11,521	4.6138619
23,040	23,041	4.6139449
46,080	46,081	4.6139863
92,160	92,161	4.6140071

The $h \rightarrow 0$ limit of this (L, B) is $L/B = 4.6140277$; a verification at $m = 184,320$ ($S_* = 4.6140174$) is running at this writing.

5 The frontier map, with rigorous walls

The budget-17 feasibility boundary was mapped around the certificate. “Infeasible” below means: over a subset of harvested constraint cells, the LP already requires total weight > 17 — a rigorous lower bound on the budget any full-coverage measure needs.

target	verdict	bound
$(L, B) = (4.613105, 0.9998)$	certified	this release
$L = 4.61311, B = 0.9998$	certified measure itself breaks	float min 0.947
$L = 4.61312, B = 0.9998$	infeasible	min-total 17.040 (2,570-position pool)
$B = 0.99975, L = 4.61306$	infeasible	min-total 17.048
$B = 0.9997, L = 4.61306$	infeasible	min-total 17.050

For completeness, three natural ideas fail and are recorded here: (i) *threshold/bundle atoms* (Levy’s $\lfloor |S|/k \rfloor$ mechanism) give $\max\text{-}\lambda = 0.987 < 1$ with 400 pair-bundles at $L = 4.61312$ — post-hoc bundling of a converged measure collapses coverage; (ii) *heterogeneous- B* atoms are dead on arrival: the capture lemma’s budget accounting (each atom captures at most one disjoint core) requires atom side $\leq$ core side, which makes coverage strictly harder; (iii) *D_4 -symmetrizing* the measure breaks capture at direction 895 (0.999565175): the D_4 orbit of a net direction leaves the verified range. All three engines reproduced that exact value independently — a useful four-way consistency check.

6 Verification

Everything in the proof is exact rational arithmetic; no floating-point step is part of the proof.

engine	code/arithmetic	result
1. difference-array engine	exact fractions, integer 2-D difference arrays	PASS, all five nets
2. Mira-acc segment-tree C++	Boost cpp_int, segment tree	EXACT_CERTIFICATE_VALID
3. Mira-acc direct-prefix C++	same file, <code>-DDIRECT_PREFIX_CHECK</code>	identical summary line
4. independent prefix engine	new Python code, per-slab prefix + exact SAT	agrees on samples

Engines 2–3 are **Mira-acc’s own exact kernel** [4], recompiled unmodified except for the removal of its D_4 -symmetry consistency assert (the assert only checks that a certificate is D_4 -symmetric; the sweep uses the atoms directly for every direction and never mirrors the measure). Both accumulation variants print, for all 92,161 directions:

```
EXACT_CERTIFICATE_VALID atoms 387 directions 92161 total_mass 16999900220/1000000000 minimum
1000222750/1000000000 slabs 68611921
```

Each engine also re-validates the global gates: total weight < 17 , strict containment $B^2(1+h)^2 < 1+h^2$, and net coverage of $[0, \pi/4]$. Full logs and the certificate ship with this release (`verification/`, `VERIFY.md`).

7 Conclusion

The certified bracket is now

$$4.6140071 < s(17) \leq 4.67553009360455\dots,$$

with the lower bound improved from 4.613028635886 to 4.6140071 ($+9.8 \times 10^{-4}$) by re-optimizing the incumbent’s measure along the (B, m) axes and by mapping the budget-17 frontier with rigorous walls. The lower-bound program in the weighted-atom framework at budget 17 is now exhausted at $(L, B) = (4.613105, 0.9998)$: both neighbouring lattice points ($L \geq 4.61312$, $B \leq 0.99975$) provably require total weight > 17 . Further progress needs either a new measure family (beyond uniform-core weighted atoms) or progress on the 28-year-old upper bound.

References

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